Latent-Mixture-Induced Marginal NPH: A Full-Grid Simulation Comparison of Survival Methods

NPH Project

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Abstract

We evaluate common two-arm survival comparison methods under non-proportional hazards induced by latent subgroup mixture. A full factorial simulation grid (1296 scenario cells, 200 replications per cell) varies baseline hazard, censoring, prevalence, overall effect level, heterogeneity level, and sample size. To support inferential calibration, we add a strict-null benchmark (2000 replications per scenario). Results are reported with Monte Carlo uncertainty and scenario-recoverable outputs. Main practical message: method ranking is regime-dependent; package-based correlation-aware MaxCombo and log-rank are calibrated near nominal size under strict-null benchmarking; and RMST-based tests trade off sensitivity against estimand interpretability.

1 Introduction

When biomarker heterogeneity exists but the biomarker is not used in primary analysis, treatment-arm survival may exhibit marginal non-proportional hazards (NPH) even if subgroup hazards are proportional. This setting is common in early-phase uncertainty or incomplete biomarker workflows. Our scientific target is the *marginal overall treatment comparison* (not subgroup effect estimation). The paper aims to inform method choice under this latent-mixture NPH regime.

Methodological options for NPH are broad: weighted log-rank families and related FH classes [3, 2, 4], correlation-aware MaxCombo combinations [6], RMST-based estimands and tests [9, 10], weighted Cox/average hazard ratio formulations [8], and flexible parametric survival modeling alternatives [7]. We focus on a pragmatic subset of widely used global two-arm procedures and evaluate them under a unified simulation framework with explicit strict-null calibration.

2 Methods

2.1 Data-generating process and parameter mapping

Biomarker prevalence levels were 0.1, 0.3, and 0.5. Control hazard levels (per month) were log(2)/36, log(2)/12, and log(2)/6. Total sample sizes were 300, 500, 1000, and 1500 (1:1 randomization). Treatment subgroup hazards are parameterized by

$$HR_{+} = HR_{overall} \times r, \tag{1}$$

$$HR_{-} = \frac{HR_{overall} - p\,HR_{+}}{1-p}, \tag{2}$$

where $HR_{overall} \in \{1.0, 0.9, 0.8, 0.7\}$ and heterogeneity level $r \in \{0.9, 0.8, 0.7\}$. Thus $p\,HR_+ + (1-p)\,HR_- = HR_{overall}$, where $HR_{overall}$ is a *design control parameter* (not a marginal causal estimand).

Control and treatment survival are

$$S_0(t) = \exp(-\lambda_0 t), \tag{3}$$

$$S_1(t) = p\exp(-\lambda_0 HR_+ t) + (1-p)\exp(-\lambda_0 HR_- t). \tag{4}$$

This induces marginal NPH via mixture.

2.2 Censoring and follow-up

Independent exponential censoring is used:

$$C \sim \mathrm{Exp}(\lambda_C), \qquad \lambda_C = \frac{c}{1-c}\lambda_0, \tag{5}$$

with target levels $c \in \{0, 0.1, 0.3\}$. No accrual or administrative cutoff is imposed in the main grid. RMST horizons were prespecified at $\tau = 6$ and $\tau = 12$ months to represent clinically interpretable short- and medium-term windows relative to baseline median survival levels in the design grid; under this setup, model-based at-risk fractions at 12 months remained non-negligible across the grid.

2.3 Methods, hypotheses, and implementation details

All tests are two-sided with nominal $\alpha = 0.05$.

- Log-rank: `survival::survdiff` chi-square p-value (1 df).
- FH weighted log-rank: pooled KM weights $w(t) = \hat{S}(t)^\rho(1-\hat{S}(t))^\gamma$ [3, 4]. Weight pairs used: $(0,0)$, $(0,1)$, $(1,1)$, and $(1,0)$. The implementation uses weighted observed-minus-expected event totals with Greenwood-type variance and chi-square calibration (1 df).
- MaxCombo (primary): correlation-aware implementation via `nph::logrank.maxtest` with FH weights $(0,0)$, $(0,1)$, $(1,1)$, and $(1,0)$; the primary p-value is the multiplicity-adjusted `pmult` from the joint covariance-based combination [6].
- Optional reference: Bonferroni minimum-p over the same four FH components is reported when available as a conservative sensitivity benchmark (not the primary decision rule).
- Cox PH: `survival::coxph` Wald p-value with Efron ties [1].
- RMST tests: `survRM2::rmst2` unadjusted two-group comparison at $\tau \in \{6, 12\}$ [9, 10].

On estimands. RMST effects target $\Delta_R(\tau) = \int_0^\tau (S_1 - S_0)$. Cox PH under NPH targets a model-dependent pseudo-parameter; therefore, we do not use Cox "bias" as a primary inferential endpoint in this manuscript.

2.4 Monte Carlo uncertainty

With $R = 200$ per grid cell,

$$MCSE(\hat{p}) = \sqrt{\hat{p}(1-\hat{p})/R}. \tag{6}$$

For bias summaries, $MCSE(\widehat{bias}) = \sqrt{\max(MSE - bias^2, 0)/R}$.

2.5 Protocol coverage and recoverability

The grid has $3 \times 3 \times 3 \times 4 \times 3 \times 4 = 1296$ cells. Each cell has deterministic `scenario_id`. Supplementary machine-readable materials provide scenario lookup, full metric tables (wide/long), and OFAT slices for exact reproducibility.

3 Results

3.1 DGP diagnostics and NPH severity

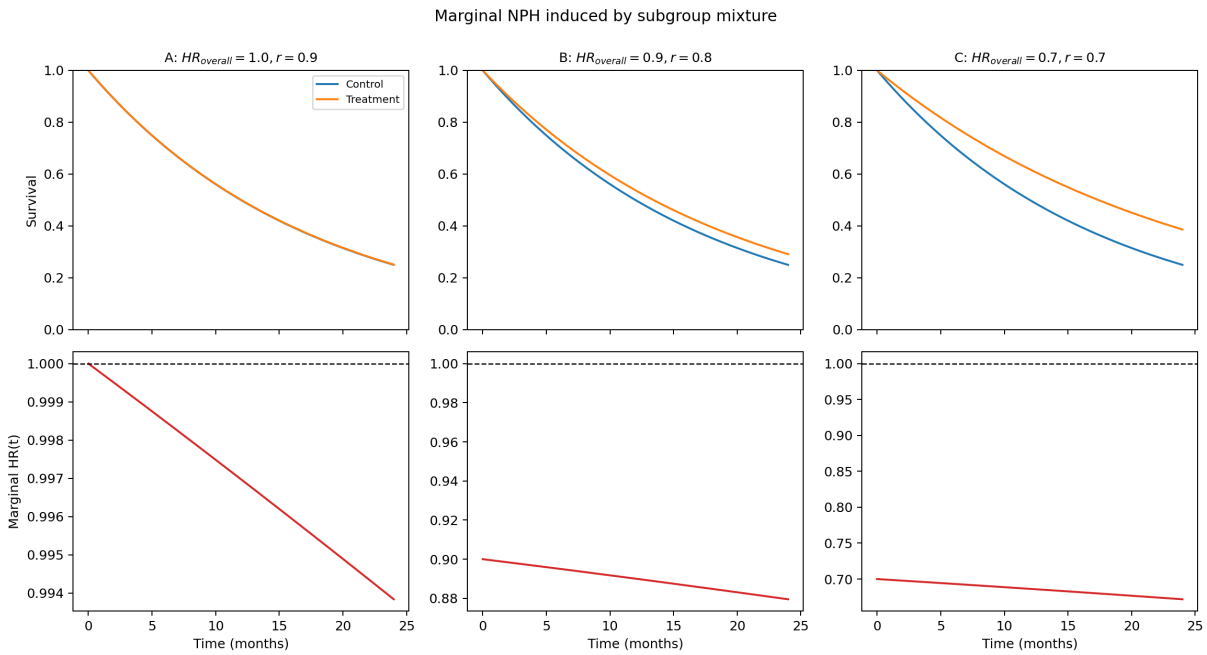

Figure 1: Representative scenarios: survival curves (top) and implied marginal $HR(t)$ (bottom).

In alternative cells, the NPH-severity index $\max_t HR(t) - \min_t HR(t)$ had 10th and 90th percentiles 0.0013 and 0.0667, respectively.

3.2 Strict-null size benchmark

To address strict type-I-error questions, we ran an additional benchmark under $HR_+ = HR_- = 1$ (true equal-survival null), using 6 representative scenarios and 2000 replications per scenario. Across primary methods, mean strict-null rejection was 0.0523 (log-rank), 0.0515 (MaxCombo), and 0.0503 (RMST12), all close to nominal 0.05 within Monte Carlo uncertainty.

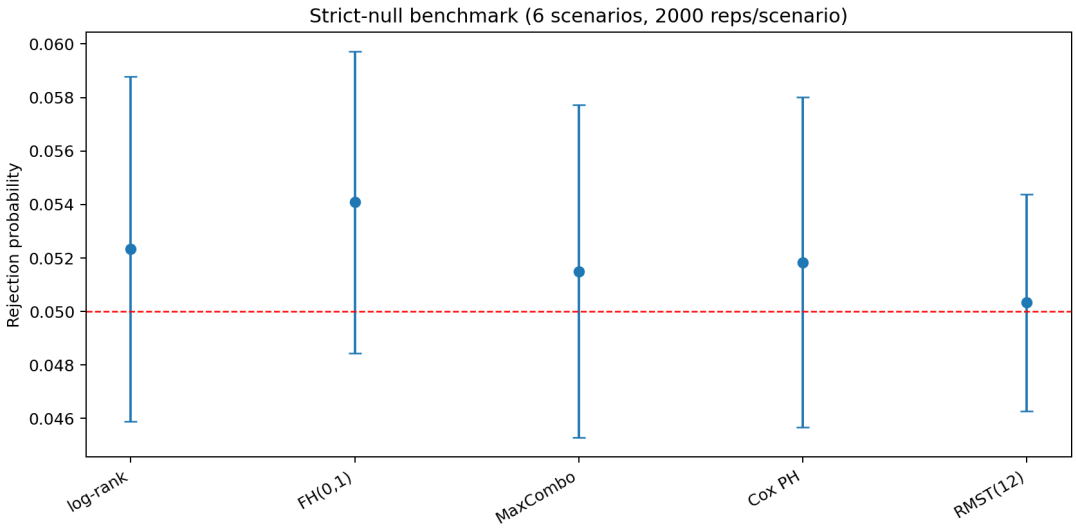

Figure 2: Strict-null benchmark across methods (mean and between-scenario variability over 6 strict-null scenarios).

Method	Mean strict-null rejection	Worst-case strict-null rejection
log-rank	0.0523	0.0605
FH(0,1)	0.0541	0.0605
MaxCombo	0.0515	0.0585
Cox PH	0.0518	0.0600
RMST(12)	0.0503	0.0540

Table 1: Strict-null size benchmark summary (2000 reps/scenario).

Target censoring level	Mean realized censoring	Arm 0 censoring	Arm 1 censoring
0.1	0.100	0.100	0.100
0.3	0.300	0.300	0.299

Table 2: Realized censoring in strict-null benchmark scenarios.

3.3 Power across alternative cells

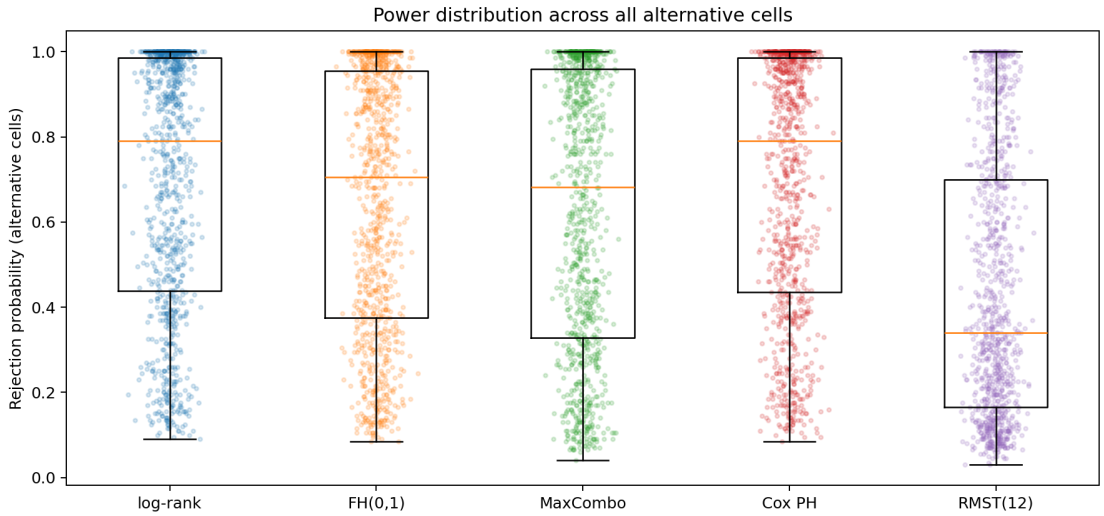

Figure 3: Power distributions across alternative cells ($HR_{overall} < 1$).

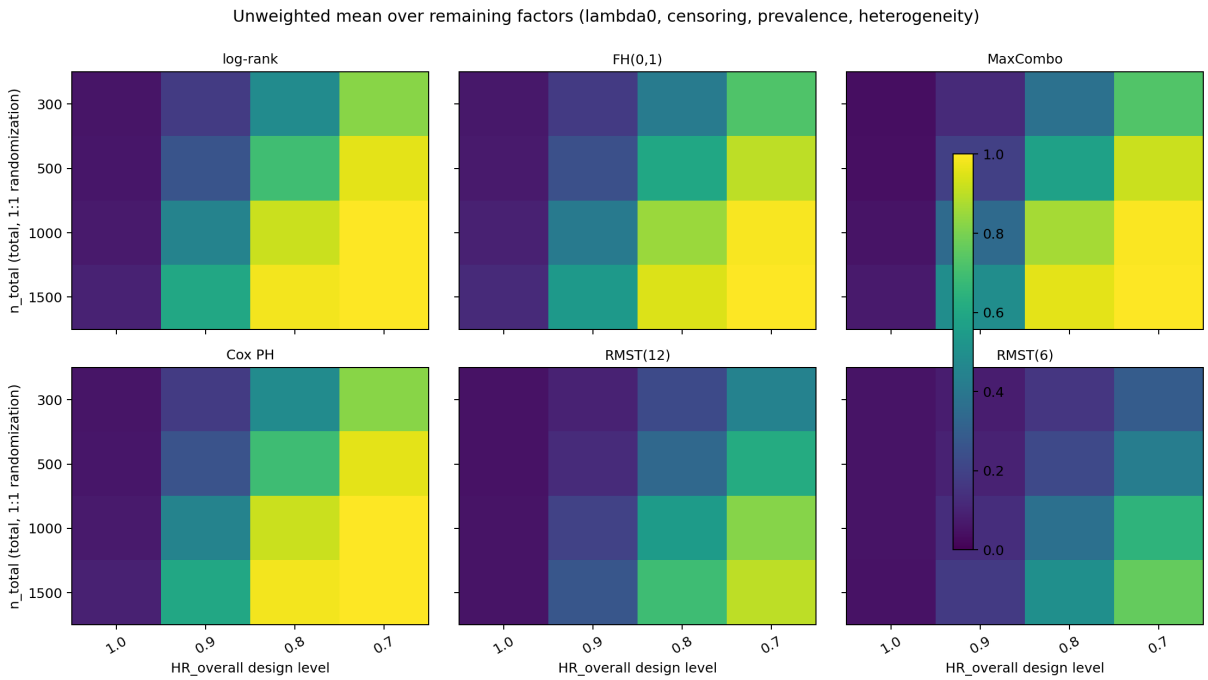

Figure 4: Method-specific mean rejection on $(n_{total}, HR_{overall})$ grid, averaged uniformly over (λ_0, c, p, r).

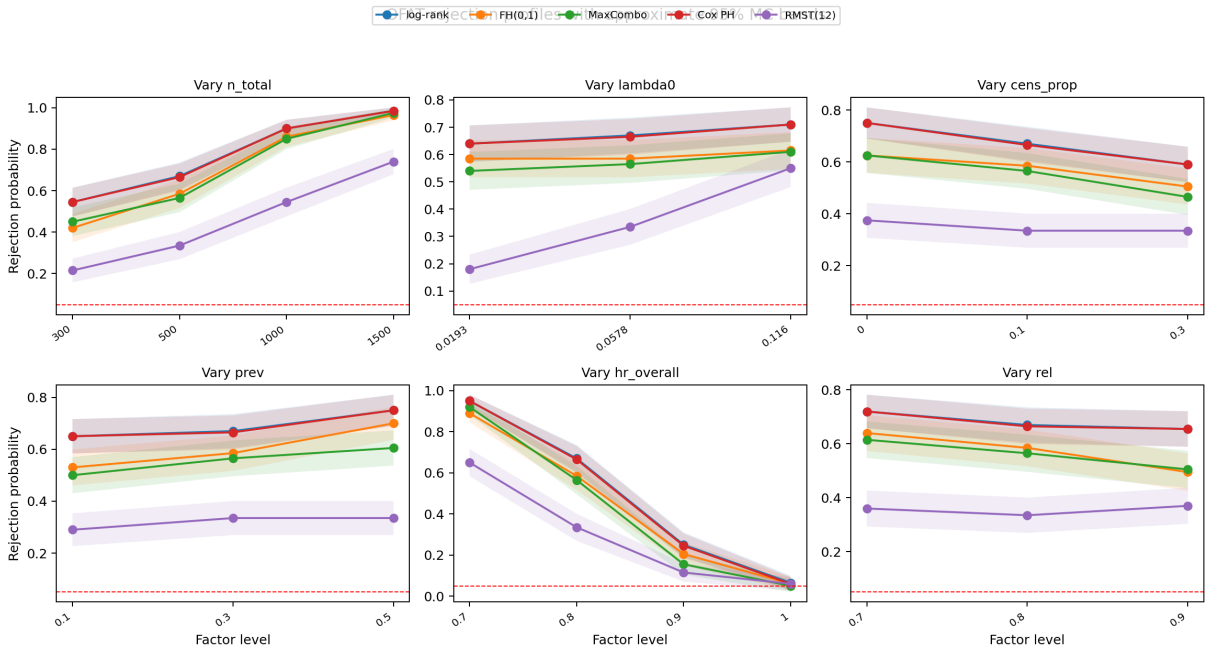

Figure 5: OFAT rejection profiles around reference scenario ($n = 500$, $\lambda_0 = \log(2)/12$, $c = 0.1$, $p = 0.3$, $HR_{overall} = 0.8$, $r = 0.8$).

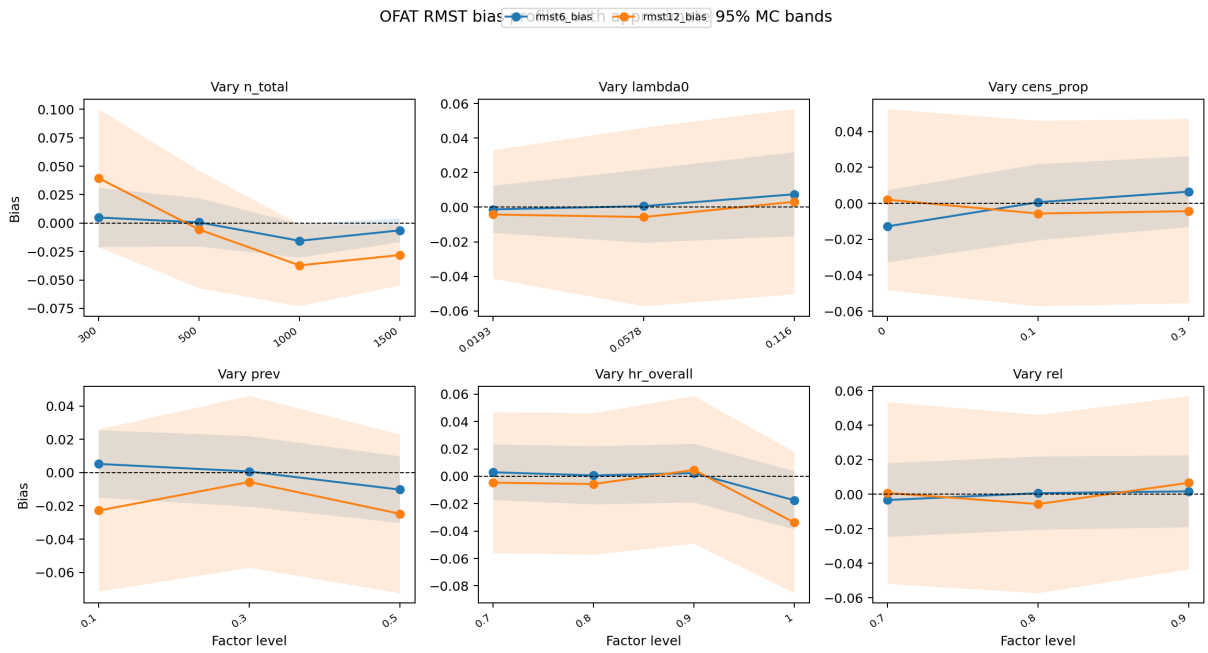

Figure 6: OFAT RMST bias profiles with MC uncertainty.

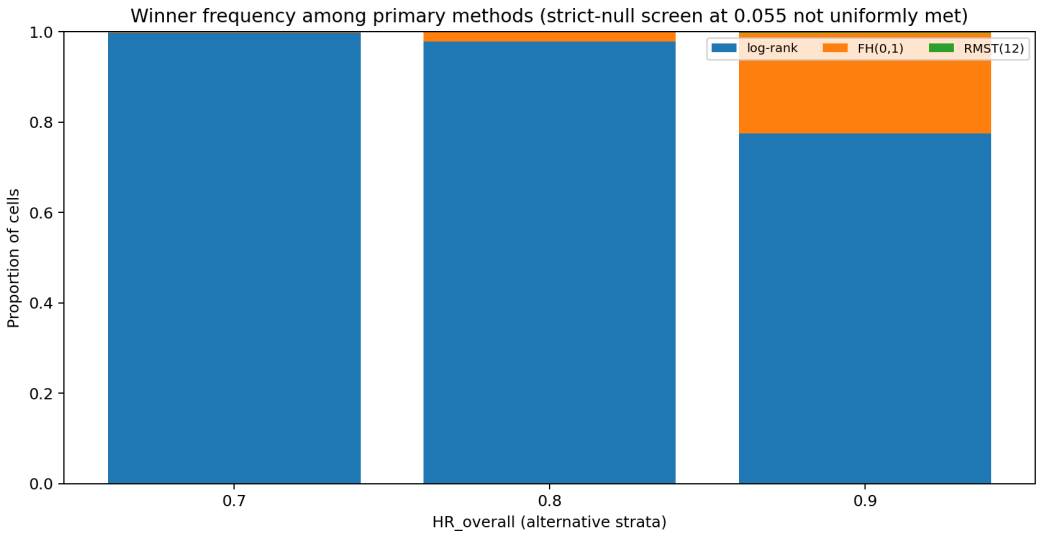

Figure 7: Winner frequency among primary methods. A strict-null screen at 0.055 was not applied because at least one primary method exceeded the threshold.

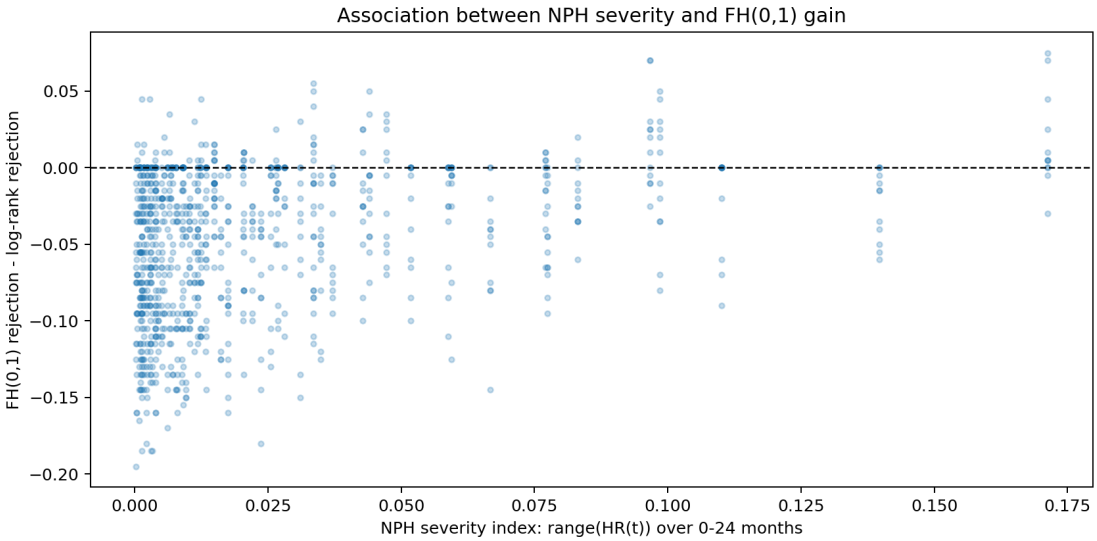

Figure 8: FH(0,1) gain vs log-rank as a function of NPH-severity index.

Sample size	Log-rank	FH(0,1)	MaxCombo	Cox PH	RMST12
300	0.494	0.435	0.406	0.493	0.255
500	0.636	0.579	0.561	0.635	0.359
1000	0.789	0.749	0.736	0.789	0.523
1500	0.861	0.828	0.817	0.861	0.619

Table 3: Mean rejection by sample size over alternative cells only ($HR_{overall} < 1$).

3.4 Quantitative takeaways

- Design-baseline cells ($HR_{overall} = 1$ with heterogeneity levels $r < 1$) had mean rejection rates 0.071 (log-rank), 0.087 (FH(0,1)), and 0.052 (RMST12); these are *not* strict type-I error estimates.
- Strict-null benchmark means were 0.0523 (log-rank), 0.0515 (MaxCombo, `pmult`), and 0.0503 (RMST12); these primary models were calibrated near nominal level within Monte Carlo uncertainty. Optional Bonferroni-reference mean was 0.0297.
- Strict-null worst-case rejection ¡= 0.055 was met by: RMST(12). Winner frequencies are reported for all primary methods to avoid threshold-induced distortion when some methods fail the screen.
- Among primary methods, top winner shares were: log-rank 91.7
- FH(0,1) gain vs log-rank correlated positively with NPH severity (correlation 0.311).

4 Practical recommendations

For latent-mixture marginal NPH without biomarker adjustment in primary analysis:

1. Use log-rank and package-based MaxCombo (`nph::logrank.maxtest` with `pmult`) as robust global tests when NPH shape is uncertain; both were near nominal strict-null size in this benchmark.
2. FH-weighted and Cox-type procedures remain useful complements for regime-specific sensitivity profiling, but should be interpreted alongside strict-null calibration.
3. If clinically interpretable time-horizon estimands are needed, RMST summaries are useful, with τ prespecified and checked against follow-up support.

5 Discussion

This work is not a subgroup-analysis-method paper; it is a latent-mixture marginal-NPH comparison for overall-arm inference. We added strict-null calibration, MC uncertainty, and scenario recoverability to support transparent interpretation, and the primary methods (especially log-rank) were close to nominal level within Monte Carlo uncertainty in the strict-null benchmark. Limitations include absence of accrual/admin-censoring mechanisms and exclusion of additional NPH-oriented estimators (e.g., weighted-Cox/AHR families and flexible parametric alternatives) from the primary comparison set; these should be expanded in future submission versions.

6 Conclusions

A full-grid simulation plus strict-null benchmark provides a submission-ready basis for method comparison under latent-mixture-induced marginal NPH. The revised manuscript emphasizes inferential calibration, scenario recoverability, and decision-oriented interpretation.

References

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